COSC3126 Data Mining
Introduction

Master of AI

SSET, RMIT University

Slides adapted from Prof. Xiaodong Li

Semester 1, 2026

What is Data Mining?

Data mining is a process of extracting and discovering patterns in large data sets involving methods at the intersection of machine learning, statistics, and database systems.

Information about the course

- Topics
  - Week 1: Introduction
  - Week 2: Classification
  - Week 3: Clustering
  - Week 4: Association Finding
  - Week 5: Attribute Selection, Visualisation
  - Week 6: Ensembles for Classification

    Week 7: No class (Personal development week)
  - Week 8: Neural Network 1
  - Week 9: Neural Networks 2, Deep Networks
  - Week 10: Case Study
  - Week 11: Naive Bayes, Text Mining
  - Week 12: Revision, Project Presentation
- Assignment 1 due by the end of week 6 (40%); Assignment 2 due by the end of week 10 (40%); Final assessment (project presentation) is at the end of week 12 (20%).

Course resources

- Course Guide: pre-requisites, course description, objectives, learning activities, and assessment overview.
- Canvas: all lecture/tutorial/lab materials are made available on Canvas, online discussion forums for general and assignment specific questions.
- Plagiarism issues: warnings and RMIT policies.

- Prescribed text book (available free via RMIT Library):
- Ian H. Witten, Eibe Frank and Mark A Hall. **Data Mining: Practical Machine Learning Tools and Techniques**, Morgan Kaufmann Publishers, 3rd edition, 2011.

- Weka data mining software tool: `https://www.cs.waikato.ac.nz/ml/weka/`
- Rich online resources available on Weka (just google them).

How the course is delivered?

- Lectures: there will be one (up to 2-hour) pre-recorded video lecture each week. Students are expected to watch the pre-recorded video lecture before attending lectorial/lab activities.
- Lectorial/Tutorial: there will be one 2-hour lectorial/tutorial session each week, run face to face in the classroom. Students are expected to ask questions about the lecture (like a Q-and-A session), and have attempted the tutorial questions before attending the session.
- Lab/Practical: there will be 1-hour lab session each week. Students are expected to have already attempted the exercises.
- Quizzes (optional): these are for testing your understanding of data mining concepts. You can attempt them after the lectorial. These quizzes do not account for any marks.

Assignments and Assessments

1. Assignment 1 Practical data mining (40%): there are 5 topics - classification, numeric prediction, clustering, association finding, and a presentation on ethical considerations.

2. Assignment 2 Neural networks (40%): there are two options:
   - Standard Assignment 2 (40%): there are 3 parts - classification, numeric prediction, and data mining application.
   - Alternative Assignment 2 (40%): Work on competition data from a data science site like kaggle.com (for the data mining application part).

3. Course project (20%): this is a team work assessment. There are 3 parts - Bayes network, text mining and in-class presentation.

Tools for Data Mining

	Weka	R	Python
Algorithms	Widest Range	Most of the popular ones	Most of the popular ones
Programming	None	Programming in R language	Python
Data Prep	Wide range of filters	Some support	Some support

- Familiarity with Unix scripting is helpful
- For some tasks related to transforming data there is likely a Unix program that does it.

Summary to Assignment 1

1. The process of knowledge discovery in data sets
2. Data Mining Tasks
   - Classification
   - Numeric Prediction, time series prediction
   - Association Finding
   - Clustering
   - Attribute Selection

2 Parts: Assignment 1 Part 1 - Due in week 4
Assignment 1 Part 2 - Due in week 6

Why is Data Mining Necessary?

Data comes like water out of a fire hydrant. You can't drink it (Anon)
We are drowning in information but starving for knowledge (John Naisbett)

- Hardware advances in data collection and storage have far outpaced software advances in data analysis and manipulation.
- Organizations collect more data than they can handle.
- Data that may never be analysed is still collected out of fear of missing something that may be important.
- As databases grow decision making directly from their contents is not feasible; knowledge derived from the data is needed
- Supermarket chains, credit card companies, banks routinely generate gigabytes of data daily
- Exponential growth of web pages
- Entering the era of (not tera) peta and exa bytes
- Social Media: Google, Facebook, Twitter
- FANG (Facebook, Amazon, Netflix, Google)

Knowledge Discovery & Data Mining

- *Knowledge Discovery in Databases (KDD)* is the non-trivial process of identifying valid, novel, potentially useful and ultimately understandable patterns in data [Fayyad]
- 'Golden Nuggets'
- *Data mining* is the part of the KDD process relating to methods for extracting patterns from data [Fayyad]
- *Data mining* is a problem solving methodology that finds a logical or mathematical description, of a complex nature, of patterns and regularities in a set of data [Decker and Focardi]
- In some current usage KDD = Data Mining
- Knowledge extraction, information discovery, information harvesting, data archaeology, data dredging, data pattern processing
- Emerging new names: 'Data Analytics', 'Predictive Analytics', 'Big Data', 'Data Science'.

Examples (1)

- Fraudulent credit card transactions
- Good/bad loan risks
- New class of stars
- Put perfume and greeting cards together and you'll sell more of each
- People who bought this product/book also bought these other products/books
- Recognition of specific market segments that respond to specific characteristics
- Telephone customers who will switch to another provider
- Fraudulent international telephone calls
- Ineffective advertising

Examples (2)

- Dating Site: both men and women's actual behavior differs significantly from their stated tastes and preferences which they outline when they first sign up. In other words, people are not as fussy about partners as they make out.

  http://www.technologyreview.com/view/524081/data-mining-reveals -the-surprising-behavior-of-users-of-dating-websites/

- What goods should be promoted to this customer?
- http://www.researchpipeline.com/wordpress/category/data-mining/
- The Economist recently had an interesting article on how insurance companies are increasingly using data mining to "analyze risk." That is, they look through the data which was originally collected for the purpose of better marketing, and use it as a tool to see if you lead an unhealthy life. However, the really interesting point is highlighted by Kashmir Hill, where an exec at a datamining company admits that he's changed his habits because of this. Not his eating habits, mind you. But how he purchases food.

Examples (3)

- **Target** has been highlighted for its uncanny ability to predict when women shoppers are pregnant. Public birth records just aren't updated anywhere near fast enough for retailers who want to know when to start sending targeted ads to new parents ASAP (But just wait. We'll be sending you coupons for things you want before you even know you want them?).
- Marissa Mayer said credit card companies can predict a divorce with 98% accuracy two years before it happens. Considering 50% of marriages end in divorce anyway, that might not be considered impressive.
- Data Mining applications in health care.
- Data Crunchers who helped Obama win:

  http://swampland.time.com/2012/11/07/inside-the-secret-world-of-quants-and-data-crunchers-who-helped-obama-win/

More data mining examples can be found in Witten, et al. (2011) Sections 1.2 & 1.3.

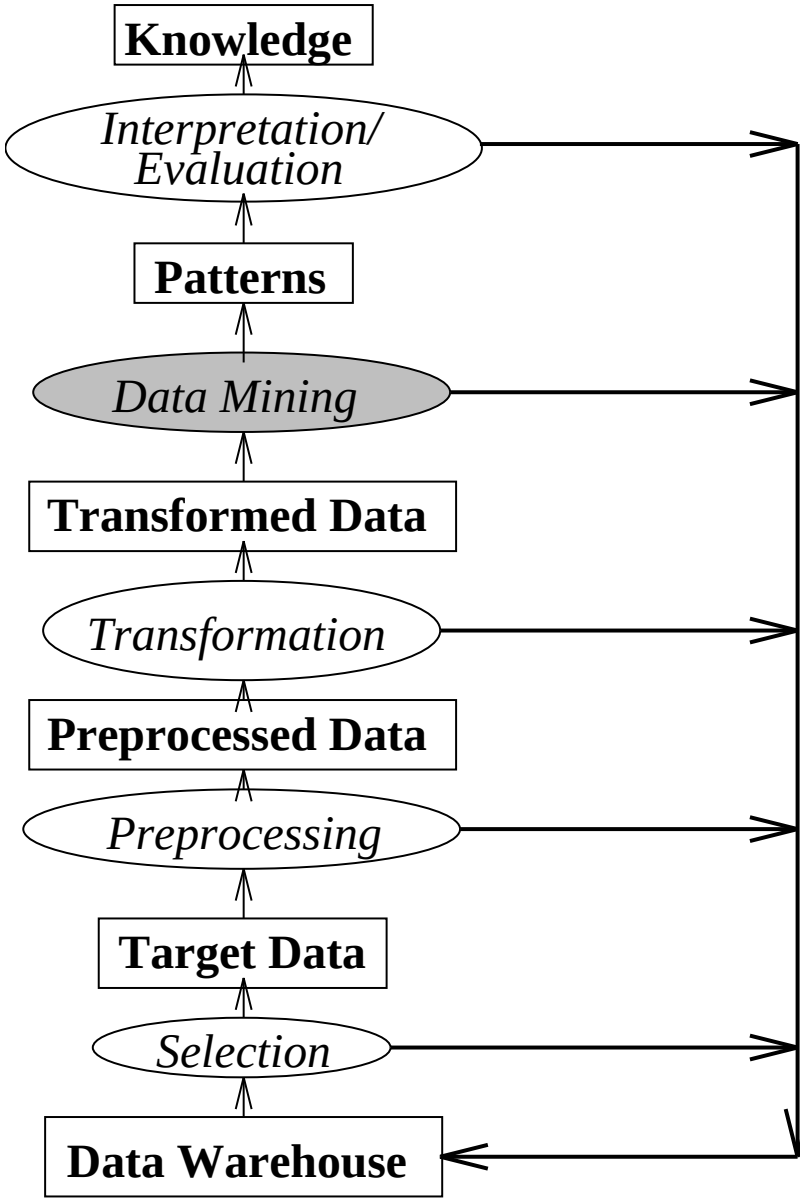
Data Mining and Ethics

Data mining practitioners need to be aware of the ethical issues that surround their data mining applications.

- Using gender and racial information for medical diagnosis is ok, but not for using the same information for mining loan payment behaviour.
- Difficulty in anonymizing data: over 85% of Americans can be identified from publicly available records using just three pieces of information: five-digit zip code, birth date, and gender.

Several more examples can be found in Section 1.6 of Witten et al. (2011).

Knowledge Discovery in Databases (KDD)



KDD and Data Mining

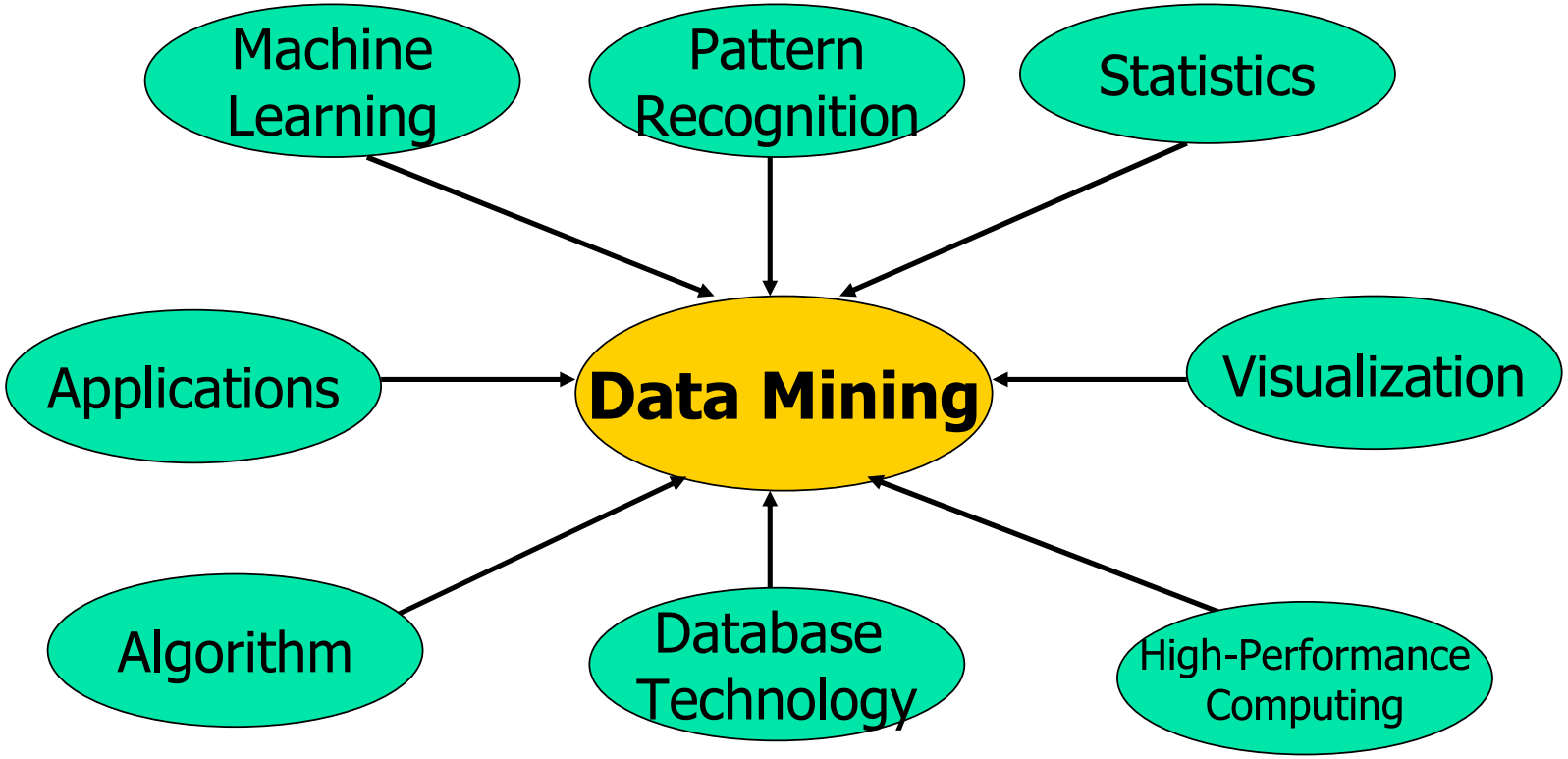
- KDD refers to the overall process of discovering useful knowledge from data, and **data mining** refers to a particular step in this process (see the previous slide).
- Data mining is the application of specific algorithms for extracting patterns from data.
- The additional steps in the KDD process, such as data preparation, data selection, data cleaning, incorporation of appropriate prior knowledge, and proper interpretation of the results of mining, are essential to ensure that useful knowledge is derived from the data.

Basic Concepts

- **Data** are a set of facts (for example, cases in a database);
- **Pattern** is an expression in some language describing a subset of the data or a model applicable to the subset.
- **Extracting a pattern** designates fitting a model to data; finding structure from data; or, in general, making any high-level description of a set of data.
- The term **process** implies that KDD comprises many steps, which involve data preparation, search for patterns, knowledge evaluation, and refinement, all repeated in multiple iterations.

Interdisciplinary Field

Data Mining: Confluence of Multiple Disciplines



Data Mining Methods

Data mining methods can be viewed as consisting of three primary algorithmic components:

1. **model representation**: the language used to describe discoverable patterns.
2. **model evaluation**: quantitative statements (or fit functions) of how well a particular pattern (a model and its parameters) meets the goals of the KDD process.
3. **search**: both parameter search and model search, i.e., finding the parameters and models from the selected family that optimize the evaluation criteria.

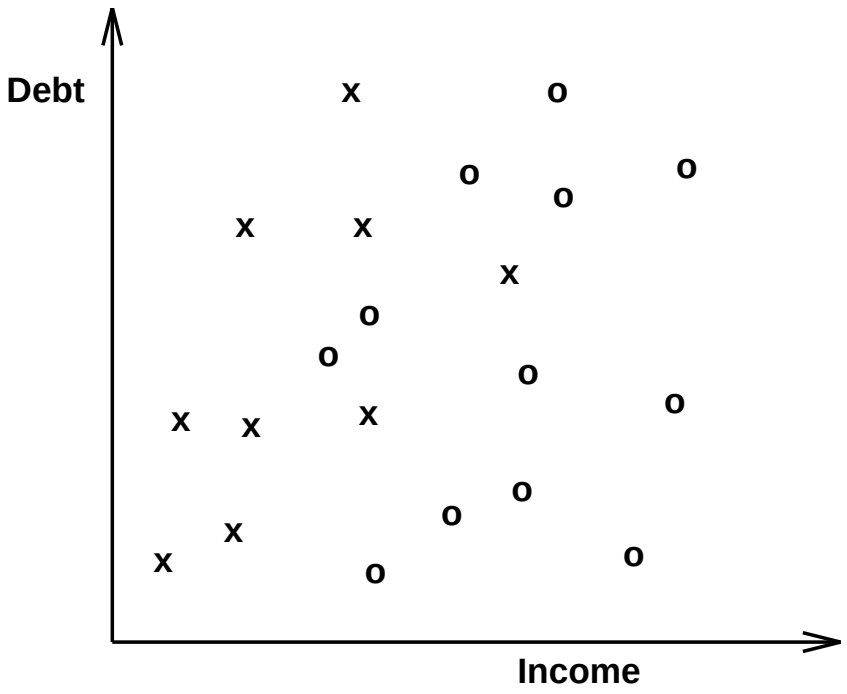
Example

Consider the data from a loan's database:

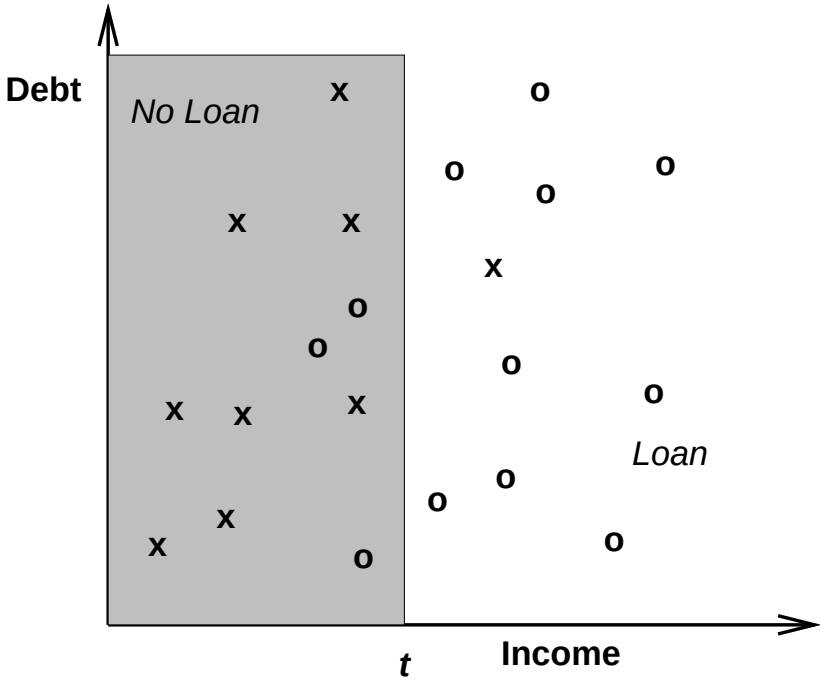
- What can be discovered? Patterns? Regularities?
- The bank would like to know whether an applicant for a loan will pay it back or not.

Income	Debt	Defaulted?
$20,000	$1,000	no (o=good)
$50,000	$25,000	yes (x=bad)
.	.	.
.	.	.

Scatter Plot



Threshold

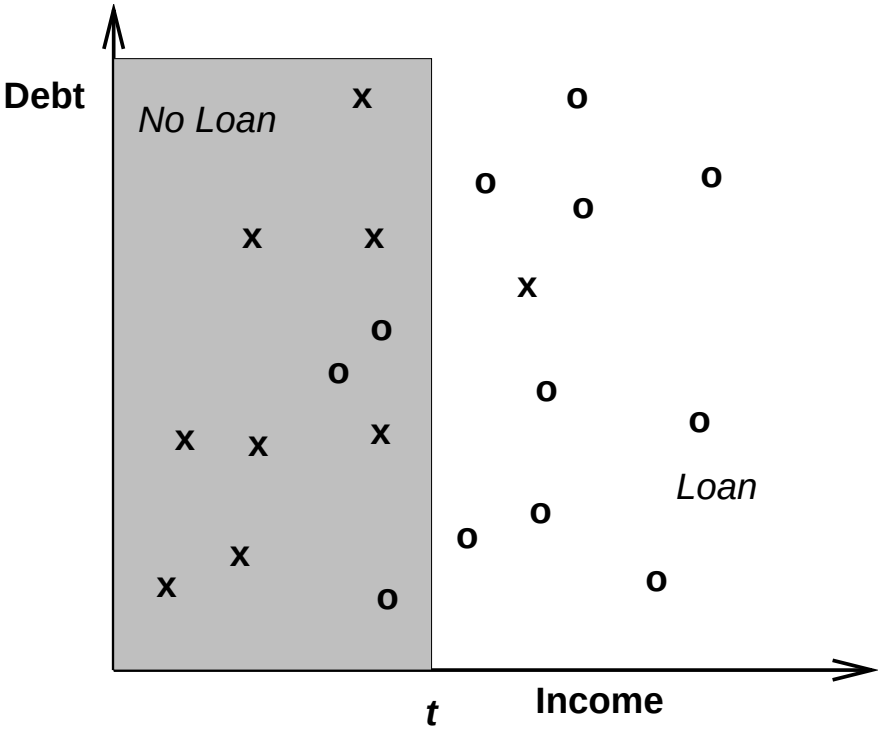


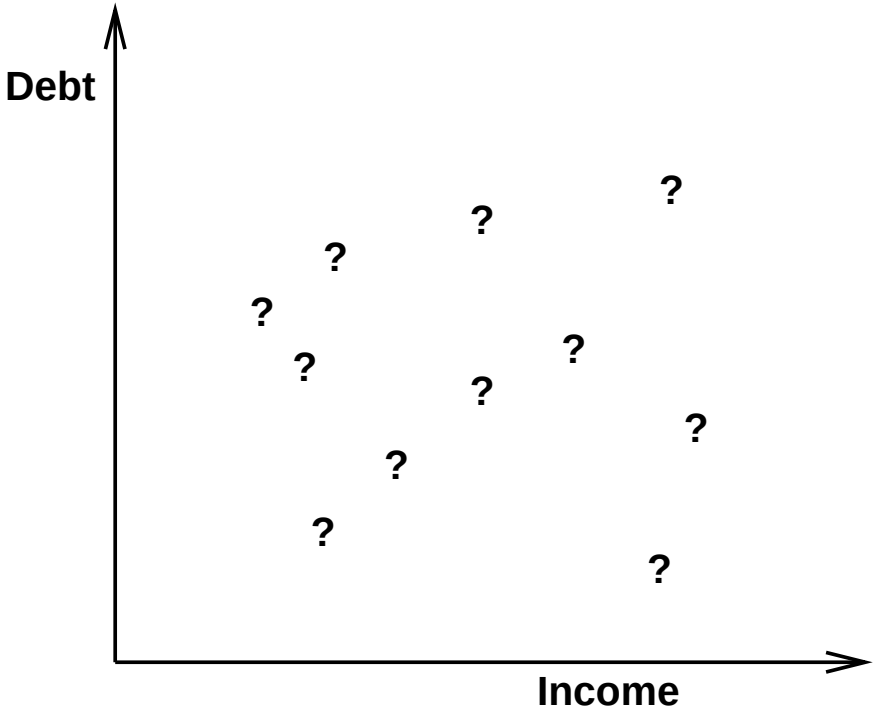
Suppose that we ‘extract’ the classification rule:

$$\text{if Income} > t \text{ then grant loan}$$

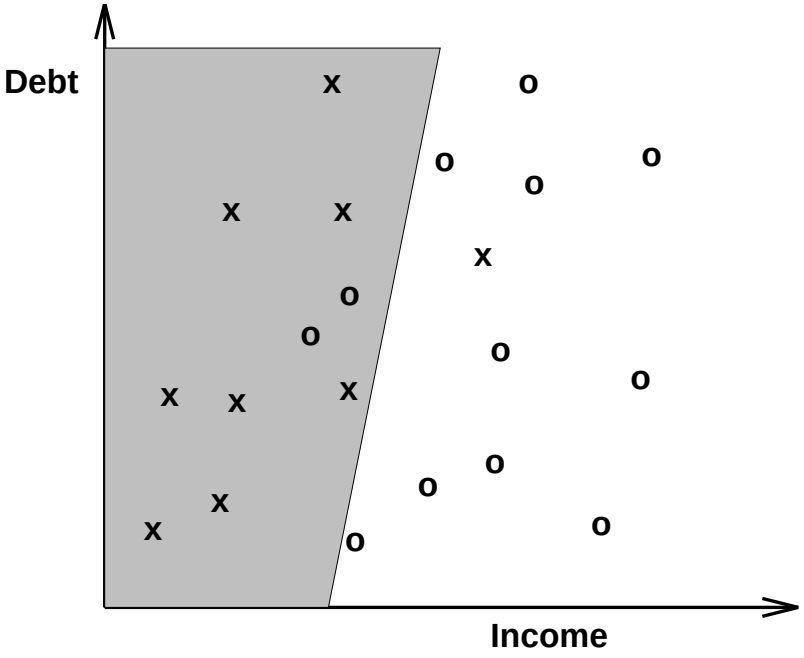
1. How many errors would we make on this data?
2. How many errors on new customers?
3. How should we choose t?
4. Can we find a better rule?

Threshold





Linear Decision Boundary



If the equation of separating line is

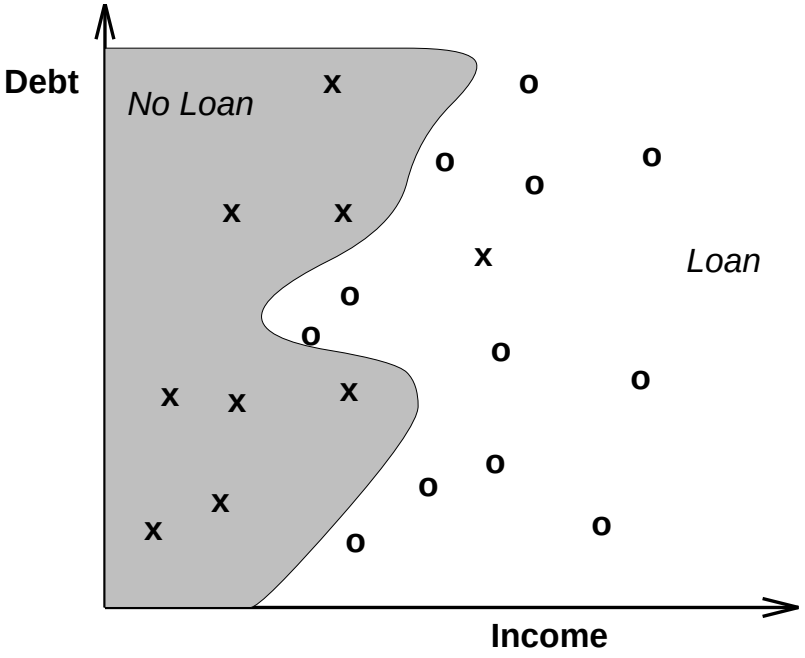
$$Debt = \alpha.income + \beta$$

we can 'extract' the classification rule:

$$if\ Debt < \alpha.income + \beta\ then\ grant\ loan$$

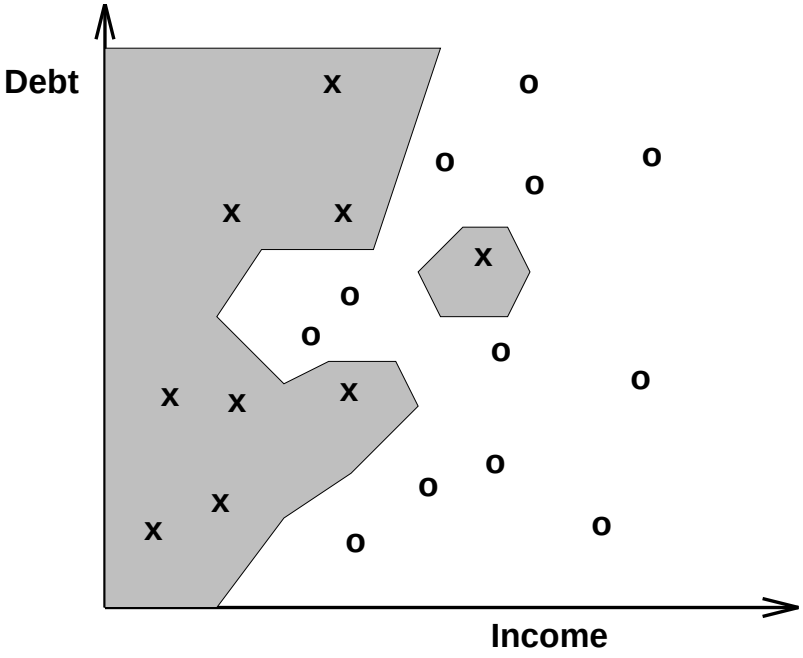
1. How many errors would we make on this data?
2. How many errors on new customers?
3. How can we find the best line?
4. Can we find a better rule?

Nonlinear Regions



1. How many errors would we make on this data?
2. How many errors on new customers?
3. How can we find the best curve? What is its equation?
4. Many classifiers will give a decision but not the equation of the separating curve.

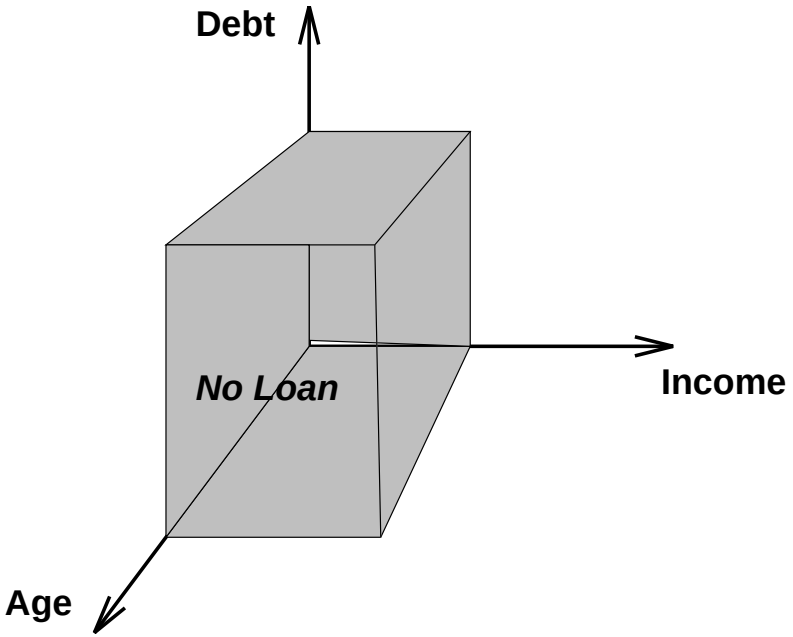
Nearest Neighbour



- Each unknown point is given the classification of its closest (k) neighbours
- How many errors on new customers?
- How can we find the best curve? What is its equation?
- Nearest neighbour can give us the regions but not the equation of the separating curve.

Higher Dimensions

- Suppose we believe that older people are more likely to pay off loans than younger people. We can include age in the decision.



- If we also include bank balance how can we visualize the result?
- How can we include male vs female and other non numeric data?

Classifiers

- A *Classifier* is a procedure which takes as input data about a particular case and outputs one of a set of predefined labels for it.
- Example 1: Take as input debt and income, output GRANT LOAN or DONT GRANT LOAN
- Example 2: Take as input patient medical data, output SICK or NOT SICK
- Example 3: Take as input weather data, output WILL RAIN TOMORROW or WILL NOT RAIN TOMORROW
- Example 4: Take as input various attributes of mushrooms, output POISONOUS or EDIBLE
- Example 5: Take as input some measurements of the iris flower, output SETOSA, VIRGINICA, VERSICOLOR
- Note: There can be more than 2 classes

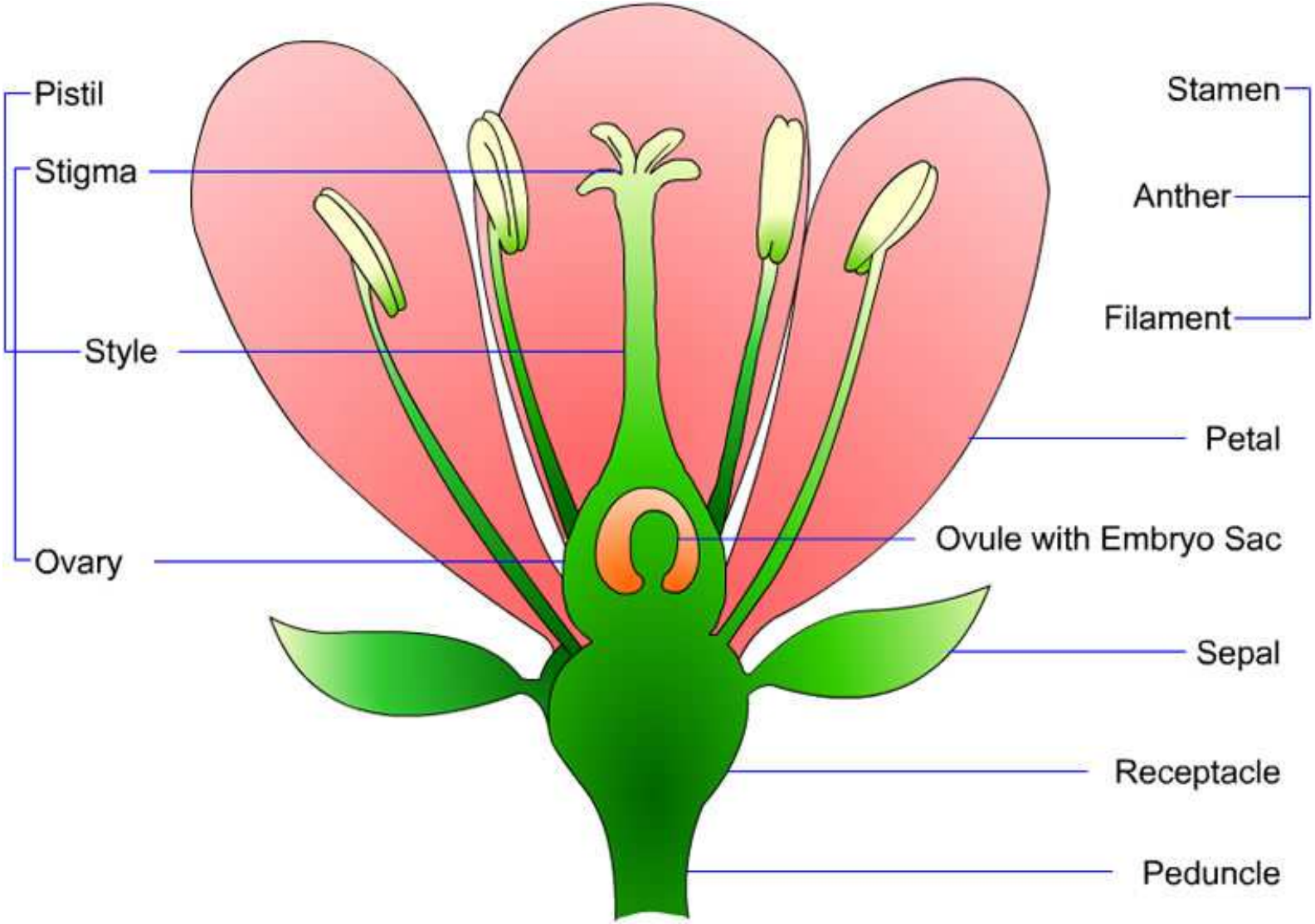
Learning of Classifiers from Data

- There are literally hundreds of algorithms for doing this
- Many are available in the WEKA package
- Basic classifiers [easy to understand the algorithm]:
  - Nearest Neighbour
  - OneR
  - ZeroR
- A bit more complex
  - Decision Tree
  - Naive Bayes
- Most Complex
  - SVM (Support Vector Machine)
  - Neural Networks

No Free Lunch Theorem

- There is no one best classification algorithm over all data sets
- Experience has shown that support vector machine (SVM) algorithms are generally a bit more accurate on many data sets.
- Recently XGBOOST seems to have the edge.

Iris Flower


Pistil
Stigma
Style
Ovary
Stamen
Anther
Filament
Petal
Ovule with Embryo Sac
Sepal
Receptacle
Peduncle

ARFF File of Iris Data

[ARFF: Attribute-Relation File Format)]

```
@RELATION iris
% R.A. Fisher, 1936
@ATTRIBUTE sepallength  REAL
@ATTRIBUTE sepalwidth   REAL
@ATTRIBUTE petallength  REAL
@ATTRIBUTE petalwidth   REAL
@ATTRIBUTE class        {Iris-setosa,
                        Iris-versicolor,
                        Iris-virginica}

@DATA
5.1,3.5,1.4,0.2,Iris-setosa
4.9,3.0,1.4,0.2,Iris-setosa
7.0,3.2,4.7,1.4,Iris-versicolor
6.4,3.2,4.5,1.5,Iris-versicolor
6.9,3.1,4.9,1.5,Iris-versicolor
6.3,3.3,6.0,2.5,Iris-virginica
5.8,2.7,5.1,1.9,Iris-virginica
7.1,3.0,5.9,2.1,Iris-virginica
.
.
```

50 examples of each class, 150 in total.

Nearest Neighbour Classifier

- Training algorithm
  - None
- Deployment stage (Testing)
  - Load the training data into memory
  - Load the unseen example
  - For each training example

    Compute distance to unseen example

    $$||\vec{x} - \vec{y}|| = \sqrt{\sum_{i=1}^{m} (x_i - y_i)^2}$$

    m is the number of attributes
  - Return the class of the example with shortest distance

WEKA IB1

```
=== Run information ===
Scheme:weka.classifiers.lazy.IB1
Relation: iris
Instances:150
Attributes:5
              sepallength
              sepalwidth
              petallength
              petalwidth
              class
Test mode:split 66.0% train, remainder test

=== Classifier model (full training set) ===

IB1 classifier

Time taken to build model: 0.02seconds

=== Evaluation on test split ===
=== Summary ===

Correctly Classified Instances          49               96.0784 %
Incorrectly Classified Instances         2                3.9216 %
Total Number of Instances               51

=== Confusion Matrix ===

  a  b  c   <-- classified as
 15  0  0 |  a = Iris-setosa
  0 19  0 |  b = Iris-versicolor
  0  2 15 |  c = Iris-virginica
```

The OneR (1R) Classifier

```
For each attribute,
   For each value of the attribute, make a rule as follows:
      count how often each class appears
      find the most frequent class
      the rule assigns that class to this attribute-value
   Calculate the error rate of the rules
Choose the rules with the smallest error rate
```

OneR Example

	Age	Prescription	Astigmatic	Tears	Lenses
1	8	myope	no	reduced	NO
2	12	myope	yes	normal	HARD
3	10	hypermetrope	no	normal	SOFT
4	14	hypermetrope	yes	reduced	NO
5	15	myope	no	normal	SOFT
6	24	myope	yes	reduced	NO
7	35	hypermetrope	yes	reduced	NO
8	55	hypermetrope	yes	normal	NO
9	56	myope	no	normal	NO
10	60	myope	yes	normal	HARD
11	65	hypermetrope	no	reduced	NO
12	72	hypermetrope	no	normal	SOFT

```
Prescription = myope;
   Lenses = NO 3 times; = HARD 2 times; = SOFT 1 times
Prescription = hypermetrope;
   Lenses = NO 4 times; = HARD 0 times; = SOFT 2 times

Rule for Prescription Attribute

If Prescription = myope then Lens = NO
If Prescription = hypermetrope then Lenses = NO

This rule will be correct 7  times and wrong 5 times on the training data.

Best Rule (Weka format)

Tears:
        reduced -> NO
        normal  -> SOFT
(8/12 instances correct)
```

MUSHROOM DATA

```
@relation mushroom
@attribute 'cap-shape' { 'b', 'c', 'f', 'k', 's', 'x'}
@attribute 'cap-surface' { 'f', 'g', 's', 'y'}
@attribute 'cap-color' { 'b', 'c', 'e', 'g', 'n', 'p', 'r', 'u', 'w', 'y'}
@attribute 'bruises?' { 'f', 't'}
@attribute 'odor' { 'a', 'c', 'f', 'l', 'm', 'n', 'p', 's', 'y'}
@attribute 'gill-attachment' { 'a', 'd', 'f', 'n'}
@attribute 'gill-spacing' { 'c', 'd', 'w'}
@attribute 'gill-size' { 'b', 'n'}
@attribute 'gill-color' { 'b', 'e', 'g', 'h', 'k', 'n', 'o', 'p', 'r', 'u', 'w', 'y'}
@attribute 'stalk-shape' { 'e', 't'}
@attribute 'stalk-root' { 'b', 'c', 'e', 'r', 'u', 'z'}
@attribute 'stalk-surface-above-ring' { 'f', 'k', 's', 'y'}
@attribute 'stalk-surface-below-ring' { 'f', 'k', 's', 'y'}
@attribute 'stalk-color-above-ring' { 'b', 'c', 'e', 'g', 'n', 'o', 'p', 'w', 'y'}
@attribute 'stalk-color-below-ring' { 'b', 'c', 'e', 'g', 'n', 'o', 'p', 'w', 'y'}
@attribute 'veil-type' { 'p', 'u'}
@attribute 'veil-color' { 'n', 'o', 'w', 'y'}
@attribute 'ring-number' { 'n', 'o', 't'}
@attribute 'ring-type' { 'c', 'e', 'f', 'l', 'n', 'p', 's', 'z'}
@attribute 'spore-print-color' { 'b', 'h', 'k', 'n', 'o', 'r', 'u', 'w', 'y'}
@attribute 'population' { 'a', 'c', 'n', 's', 'v', 'y'}
@attribute 'habitat' { 'd', 'g', 'l', 'm', 'p', 'u', 'w'}
@attribute 'class' { 'e', 'p'}
@data
'x','s','n','t','p','f','c','n','k','e','e','s','s','w','w','p','w','o','p','k',
's','u','p'
'x','s','y','t','a','f','c','b','k','e','c','s','s','w','w','p','w','o','p','n',
'n','g','e'
'b','s','w','t','l','f','c','b','n','e','c','s','s','w','w','p','w','o','p','n',
'n','m','e'
.
.
8124 examples
```

OneR ON MUSHROOM DATA

```
=== Run information ===
Scheme:       weka.classifiers.rules.OneR -B 6
Relation:     mushroom
Instances:    8124
Attributes:   23

Test mode:    split 66.0% train, remainder test

=== Classifier model (full training set) ===
odor:
        a       -> e
        c       -> p
        f       -> p
        l       -> e
        m       -> p
        n       -> e
        p       -> p
        s       -> p
        y       -> p
(8004/8124 instances correct)

Time taken to build model: 0.14 seconds

=== Evaluation on test split ===
=== Summary ===
Correctly Classified Instances        2724               98.6242 %
Incorrectly Classified Instances        38                1.3758 %
Total Number of Instances             2762

=== Confusion Matrix ===
    a    b   <-- classified as
 1410    0 |    a = e
   38 1314 |    b = p
```

Readings and Acknowledgment

- Readings:

  Fayyad, U., Piatetsky-Shapiro, G. & Smyth, P. (1996). “From data mining to knowledge discovery in databases". *AI magazine*, Vol:17, pp.37 - 54.

  Witten et al. (2011): Chapter 1.

- Acknowledgment: these slides are based on the above paper by Fayyad et al. (1996), and those originally developed by Vic Ciesielski in the previous semesters.